

Artificial Intelligence Lab-(CSE4132)

1. Prepare a PDF report using any LaTeX software. In the report, design and implement a Fully Connected Feed-forward Neural Network (FCFNN) to learn and solve the linear equation:

$$y = 7x + 12$$

Create suitable training, validation, and testing datasets for this equation. Train the model using your prepared data, test its performance, and compare the actual output values with the predicted output values using a proper plot.

2. Write a PDF report using any LaTeX software where you develop an FCFNN model for the following quadratic function:

$$y = 2x^2 + 6x + 8$$

For this task, prepare separate datasets for training, validation, and testing. After training the neural network, evaluate the model with the test data and show a comparison between the original y values and the predicted y values through graphs.

3. Using any LaTeX software, create a PDF report on building a Fully Connected Feed-forward Neural Network (FCFNN) for the cubic equation:

$$y = 5x^3 + 2x^2 + 4x + 9$$

Prepare the necessary training, validation, and test data for the equation. Train the network, test its accuracy, and present the difference between the real y values and the model's predicted y values using visual plots. Also include a short discussion on how the cubic power of x may influence the model structure and the amount of data required.

4. Prepare a PDF report using any LaTeX software. In the report, design a Convolutional Neural Network (CNN) model for a 10-class classification problem. Train and test the model using the following datasets:

- i. Fashion MNIST
- ii. MNIST English
- iii. CIFAR-10

5. Write a report in pdf format using any Latex system after:

- training a binary classifier, based on the pre-trained VGG16, by transfer learning and fine tuning.
- showing the effect of fine-tuning:

- i. whole pre-trained VGG16
- ii. partial pre-trained VGG16

Microprocessor Interfacing and Microcontrollers

Lab-(CSE4142)

1. Design and implement a simple microcontroller-based output control system using an STM32 development board. Configure the project environment, set the required board settings, and write an embedded program to control a digital output device available on the board.

Generate the necessary programming file after a successful build and transfer it to the microcontroller using an external programming tool. Finally, observe the hardware response and verify whether the program works correctly.

Parallel Processing and Distributed System Lab (CSE4112)

CUDA BASED PROBLEM

1. You are given several phonebook text files. Each file contains many records, and every record is written on a separate line in the following format:

id, person_name, phone_number

Write a **CUDA-based program** that searches through all the phonebook files in parallel. The program should find every record where the phone_number contains a given substring `str`.

Your program should take the number of CUDA threads, the list of input phonebook files, and the search substring as input. It should generate an output file containing all matching records with their id, person_name, and phone_number. Also, calculate and display the total execution time.

Input: Number of CUDA threads, input phonebook text files, and substring `str`

Output: Total execution time and an output file containing all matched records in the format:

id, person_name, phone_number

2. You are given a text file containing a large number of integer values. Each integer is within the range **0 to 255**. Your task is to calculate the frequency of each value and generate a **global histogram** using CUDA.

Write a **CUDA C/C++ program** that reads the integers from an input file and processes them in parallel on the GPU. Each CUDA thread should handle one or more data elements and update the histogram array according to the value found. Since multiple threads may update the same histogram position at the same time, proper synchronization or atomic operation should be used.

The program should take the number of CUDA threads per block, the input file name, and the output file name as command-line arguments. It should display the total number of input values, number of blocks used, threads per block, and execution time. Finally, the histogram result should be written to an output file.

Input:

Number of CUDA threads per block, input text file containing integers, and output file name.

Output:

Total execution time and an output file containing the global histogram values in the format:

H[0], H[1], H[2], ..., H[255]

Note:

Only integer values from **0 to 255** should be considered for histogram calculation.

MPI BASED PROBLEM

1. You are given multiple phonebook text files, where each file stores several phonebook records. Each record is written in a separate line using the following format:

`id, person_name, phone_number`

Write an **MPI-based program** to search these phonebook files in parallel using multiple processes. The task is to find all records where the `phone_number` includes a given substring `str`.

The phonebook files should be distributed among the MPI processes so that the searching task can be performed in parallel. After completing the search, the matching records should be collected and written to an output file. The program should also report the total execution time.

Input:

Number of MPI processes, list of phonebook text files, and substring `str`

Output:

Total execution time and an output file containing the matching records in the format:

`id, person_name, phone_number`

Digital Image Processing Lab-(CSE4162)

1. Take grayscale image of size 512x512 and perform the following operations -

- (a) Decrease its spatial resolution by half.
- (b) Repeat this process every time and observe its change when displaying in the same window size.
- (c) Decrease its intensity level resolution by one bit up to reach its binary format observe its change when displaying in the same window size.
- (d) Illustrate the histogram of the image and make single threshold segmentation observed from the histogram.

2. Take grayscale image of size 512x512 and perform the following operations –

- (a) Perform the brightness enhancement of a specific range of gray levels and observe its result.
- (b) Differentiate the results of power law transform and inverse logarithmic transform.
- (c) Find the difference image between original and the image obtained by last three bits (MSB).

3. Take grayscale image of size 512x512, add some salt & pepper noise and perform the following operations –

- (a) Apply average and median spatial filters with 5x5 mask and observe their performance for noise suppression in term of PSNR.
- (b) Use different size of mask (3x3, 5x5, 7x7) with average filter for noise suppression and observe their performance in term of PSNR.
- (c) Apply harmonic and geometric mean filter on the noisy image and compare their performance with PSNR.

Object Oriented Design and Design Patterns Lab-(CSE4122)

1. Write a Python program to demonstrate the **Adapter Design Pattern**. Create a media player application where the basic media player can play only MP3 files directly. However, it should also be able to play advanced media formats such as MP4 and VLC by using an adapter class.

Your program should include:

1. A MediaPlayer interface with a play() method.
2. An AdvancedMediaPlayer interface with separate methods for playing MP4 and VLC files.
3. Separate classes for Mp4Player and VlcPlayer.
4. A MediaAdapter class that connects the basic media player with the advanced media players.
5. An AudioPlayer class that can play MP3 files directly and use the adapter for MP4 and VLC files.
6. A Main class to test the program with MP3, MP4, and VLC file formats.

The output should show which type of media file is being played.

2. Write a Python program to demonstrate the Factory Design Pattern. Create a shape drawing application where objects of different shapes are created through a factory class instead of creating them directly in the main class.

Your program should include:

1. A Shape interface with a draw() method.
2. A Circle class that implements the Shape interface.
3. A Rectangle class that implements the Shape interface.
4. A ShapeFactory class with a method that returns the required shape object based on the given shape name.
5. A Main class to test the factory by creating shape objects and calling their draw() methods.

The output should display which shape is being drawn.

3. Write a Python program to demonstrate the Observer Design Pattern. Create a subject that stores an integer state. Whenever the state of the subject changes, all attached observers should be notified automatically and display the updated value in different formats.

Your program should include:

1. An abstract Observer class with an update () method.
2. A Subject class that stores state and maintains a list of observers.
3. An attach () method to add observers to the subject.
4. A notifyAllObservers () method to update all observers when the state changes.
5. A BinaryObserver class that displays the state in binary format.
6. A Hexadecimal observer class that displays the state in hexadecimal format.
7. A main class to create the subject, attach observers, change the state, and display the output.

The output should show that when the subject state changes, all observers are updated automatically.

4. Write a Python program to demonstrate the Singleton Design Pattern. Create a class that allows only one object to be created and provides a global access point to that object.

Your program should include:

1. A SingleObject class.
2. A private static object of the same class.
3. A private constructor to prevent object creation from outside the class.
4. A public static getInstance () method to return the single object.
5. A method to display a message.
6. A Main class to access the object using getInstance () and call the message method.

The output should show that the singleton object is accessed successfully and a message is displayed.